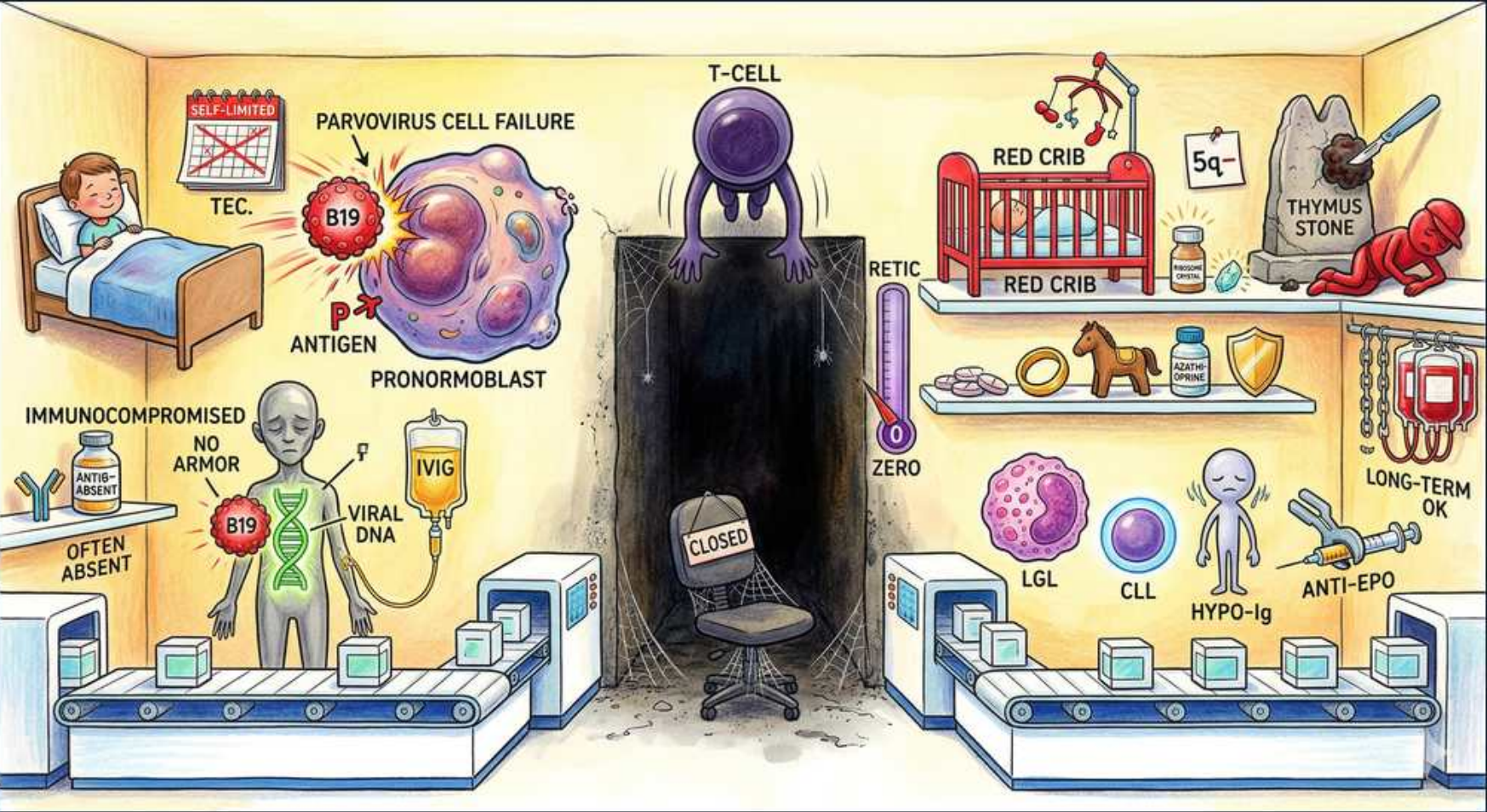


Bone Marrow Failure — The Red Cell Void (PRCA)



1. Closed red chair

WHAT YOU SEE

Empty chair with a CLOSED sign in the dark central doorway

WHAT IT ENCODES

Pure red cell aplasia is selective shutdown of erythroid production: the marrow makes white cells and platelets normally but the red-cell line is 'closed.' It presents as a severe normocytic anemia with profound reticulocytopenia and an otherwise normal CBC. The marrow shows a near-total absence of erythroid precursors with intact myeloid and megakaryocytic lineages.

BOARD TRAP

PRCA affects ONE lineage only — labeling it pancytopenia (all lineages fail) is the classic error. Normal WBC and platelets are what separate PRCA from aplastic anemia and MDS.

2. Reticulocyte zero

WHAT YOU SEE

Thermometer labeled RETIC ZERO

WHAT IT ENCODES

The hallmark lab finding is severe anemia with a near-zero reticulocyte count (reticulocytopenia), reflecting absent erythroid output, while WBC and platelet counts remain NORMAL. This isolated, profoundly hypoproliferative anemia is the diagnostic signature.

BOARD TRAP

Expecting low platelets and WBC alongside the anemia is wrong — in PRCA they are normal. If other lineages are also down, reconsider aplastic anemia or MDS rather than PRCA.

3. Absent erythroid precursors

WHAT YOU SEE

PRONORMOBLAST giant cell with CELL FAILURE label

WHAT IT ENCODES

Marrow morphology shows absent maturing erythroid precursors. When the cause is parvovirus B19, the residual finding is scattered GIANT pronormoblasts (large early erythroid cells with prominent nucleoli and viral cytopathic 'lantern' inclusions) — a near-pathognomonic clue to B19.

BOARD TRAP

Giant pronormoblasts with intranuclear inclusions = think parvovirus B19. Their presence amid otherwise absent erythroid precursors is a board-favorite pointer to the viral etiology.

4. Parvovirus B19

WHAT YOU SEE

Red B19 burst with TEC and SELF-LIMITED calendar

WHAT IT ENCODES

Parvovirus B19 is the classic infectious cause of PRCA: it directly infects and lyses erythroid progenitors, halting red-cell production. In healthy hosts this is a transient, self-limited red-cell aplasia. Diagnose chronic/immunocompromised infection by B19 DNA PCR (antibody serology may be falsely negative if the host can't mount a response).

BOARD TRAP

In patients with chronic hemolysis (sickle cell, hereditary spherocytosis), B19 precipitates a transient APLASTIC CRISIS — an abrupt hemoglobin drop with reticulocytopenia — because they cannot tolerate even a brief halt in erythropoiesis. Don't mistake the crisis for worsening hemolysis.

5. B19 antigen

WHAT YOU SEE

ANTIGEN tag on B19 near the failing cell

WHAT IT ENCODES

Parvovirus B19 enters erythroid progenitors by binding the P antigen (globoside) on their surface — the cellular receptor that explains the virus's erythroid tropism. Rare individuals lacking the P antigen are naturally resistant to B19 infection.

BOARD TRAP

The B19 receptor is the P antigen (globoside) on erythroid cells — this receptor specificity is why B19 selectively kills the red-cell line and spares WBC/platelets.

6. Immunocompromised

WHAT YOU SEE

Gray figure NO ARMOR holding B19, ANTI-B19 absent

WHAT IT ENCODES

In immunocompromised hosts (HIV/AIDS, transplant recipients, congenital immunodeficiency, chemotherapy), the inability to mount neutralizing anti-B19 antibody allows persistent viral replication and CHRONIC PRCA with ongoing transfusion-dependent anemia.

BOARD TRAP

In the immunocompromised, B19 SEROLOGY is unreliable (antibody often absent) — diagnose with B19 DNA PCR. A negative IgM/IgG does not exclude chronic parvovirus PRCA in these patients.

7. IVIG treatment

WHAT YOU SEE

IVIG bag with VIRAL DNA strand

WHAT IT ENCODES

Chronic parvovirus B19 PRCA in the immunocompromised is treated with IVIG, which supplies pooled neutralizing anti-B19 antibody to clear the virus the host cannot. Restoring immune function (e.g., ART in HIV, reducing immunosuppression) is also key.

BOARD TRAP

IVIG is the specific therapy for parvovirus PRCA — not corticosteroids/immunosuppression (which would worsen viral clearance). Match the treatment to the cause: antiviral immune support for B19 vs. immunosuppression for autoimmune/idiopathic PRCA.

8. Thymoma stone

WHAT YOU SEE

THYMUS STONE boulder at upper right

WHAT IT ENCODES

Thymoma is the classic tumor association of PRCA (though only a minority of PRCA cases have thymoma). The mechanism is autoimmune/T-cell-mediated suppression of erythropoiesis. Workup of new PRCA should include chest imaging for a thymic mass; thymectomy resolves a subset of thymoma-associated cases.

BOARD TRAP

PRCA mandates a chest CT to look for thymoma — missing the thymoma misses a potentially surgically curable cause. (Note: thymoma is also linked to myasthenia gravis and hypogammaglobulinemia.)

9. Self-limited calendar

WHAT YOU SEE

SELF-LIMITED calendar near the child in bed

WHAT IT ENCODES

In otherwise healthy children and immunocompetent adults, parvovirus B19-induced red-cell aplasia is transient and self-limited (recovers in 1-2 weeks once antibody clears the virus). Transient erythroblastopenia of childhood (TEC) is a related self-resolving acquired PRCA of young children.

BOARD TRAP

Self-limited transient PRCA (healthy host) is distinct from the chronic aplastic crisis seen when B19 hits a patient with underlying chronic hemolysis — host status determines the course.

10. Diamond-Blackfan crib

WHAT YOU SEE

RED CRIB with baby and 5q tag

WHAT IT ENCODES

The CONGENITAL form of PRCA is Diamond-Blackfan anemia, a ribosomopathy (RPS19 and other ribosomal protein genes) presenting in infancy with macrocytic anemia, ↑HbF, ↑erythrocyte ADA, and congenital anomalies (triphalangeal thumb, craniofacial). The 5q tag also flags acquired del(5q) MDS, which can present with a red-cell-predominant cytopenia responsive to lenalidomide.

BOARD TRAP

Don't confuse congenital DBA (infants, ribosomopathy, ↑HbF/↑eADA, treated with steroids) with TEC (older toddlers, normocytic, self-resolving) or with acquired del(5q) MDS in adults (lenalidomide-responsive).

11. LGL leukemia

WHAT YOU SEE

Purple LGL large granular lymphocyte

WHAT IT ENCODES

Large granular lymphocytic (LGL) leukemia is a clonal T-cell (or NK) lymphoproliferation that causes acquired PRCA (and neutropenia) via cytotoxic T-cell suppression of erythropoiesis; it is associated with rheumatoid arthritis. Diagnosis uses flow cytometry and TCR clonality; treatment is immunosuppression (methotrexate, cyclophosphamide, cyclosporine).

BOARD TRAP

LGL leukemia is a frequently-missed clonal cause of PRCA/neutropenia — look for large granular lymphocytes on smear, RA association, and TCR clonality, especially when PRCA is unexplained.

12. CLL association

WHAT YOU SEE

CLL labeled lymphocyte

WHAT IT ENCODES

Lymphoproliferative disorders — chronic lymphocytic leukemia (CLL) being the classic one — and autoimmune diseases (SLE) are recognized causes of acquired PRCA, mediated by immune suppression of erythroid precursors. Treating the underlying CLL/autoimmune disease (or immunosuppression) reverses the aplasia.

BOARD TRAP

New PRCA warrants evaluation for an underlying lymphoproliferative disorder (CLL/LGL) or autoimmune disease — PRCA can be the presenting clue to occult CLL.

13. T-cell suppressor

WHAT YOU SEE

Large purple T-CELL figure looming over the closed central doorway

WHAT IT ENCODES

The final common mechanism of most acquired PRCA is autoimmune/T-cell-mediated suppression of erythroid progenitors — cytotoxic T cells (or autoantibodies) shut down the erythroid line. This unifies the idiopathic, thymoma-associated, LGL/CLL, and autoimmune cases, and is why immunosuppression (corticosteroids, cyclosporine, with cyclophosphamide/rituximab as needed) is first-line therapy for non-viral PRCA.

BOARD TRAP

Match mechanism to therapy: T-cell-mediated (autoimmune/idiopathic/thymoma/LGL) PRCA → IMMUNOSUPPRESSION, whereas parvovirus PRCA → IVIG. Giving immunosuppression for unrecognized chronic B19 worsens it.

14. Self-limited calendar

WHAT YOU SEE

SELF-LIMITED calendar near the child in bed

WHAT IT ENCODES

In healthy children/transient cases, PRCA is self-limited; chronic hemolytics get aplastic crisis

15. Hypogammaglobulinemia

WHAT YOU SEE

Sad gray figure labeled HYPO-Ig beside the CLL/anti-EPO row

WHAT IT ENCODES

Hypogammaglobulinemia is an associated finding in PRCA: it occurs with thymoma (Good syndrome = thymoma + immunodeficiency + recurrent infections, sometimes with PRCA) and with CLL-related immune dysregulation. Low immunoglobulins also impair clearance of parvovirus B19, predisposing to chronic B19 PRCA.

BOARD TRAP

Thymoma + hypogammaglobulinemia + recurrent infections = Good syndrome — and the same patient can have PRCA. Don't overlook the immunodeficiency angle that both ties to thymoma and permits chronic B19.

16. Anti-EPO antibodies

WHAT YOU SEE

ANTI-EPO syringe at lower right

WHAT IT ENCODES

Drugs (phenytoin, azathioprine, isoniazid, mycophenolate, sulfonamides) and anti-erythropoietin antibodies cause acquired PRCA. Anti-EPO antibody PRCA is a rare but classic complication of recombinant EPO products (historically certain epoetin formulations), in which neutralizing antibodies cross-react with native EPO, producing severe transfusion-dependent anemia.

BOARD TRAP

In a patient on erythropoiesis-stimulating agents who develops worsening anemia with reticulocytopenia, suspect anti-EPO antibody PRCA — stop the ESA. Also review the medication list for culprit drugs before labeling PRCA idiopathic.